TRACE

An Input-Only Wearable Memory

Architecture, rationale, and the case for a memory
that keeps words instead of footage

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The architecture of record is a hand-drawn napkin, photographed 2026-07-18,
transcribed verbatim in Appendix A. On any conflict, the napkin outranks this document.

Abstract

TRACE is a wearable memory system built on one uncompromising invariant: raw media never persists and never leaves the device; only typed text may cross the boundary. On-device perception converts the live world into open-vocabulary, coordinate-marked, provenance-bearing text at the moment of capture; the raw pixels and audio that produced it are discarded within milliseconds. A consolidation process — deliberately modeled on biological sleep — binds the day's scattered observations into a typed evidence graph by spatial and temporal co-occurrence, attaching a calibrated confidence to every link and refusing weak ones outright. A recall engine answers unconstrained natural-language questions from this bound text alone, citing the exact observations that support each claim and refusing honestly when the answer was never perceived.

This document is the complete argument for that architecture. It derives the design from first principles (Chapters 1–2), situates it against forty years of lifelogging, egocentric-vision, and retrieval-augmented-generation research (Chapter 3), states the seven locked invariants that govern all implementation (Chapter 4), and walks the full pipeline from photons to answers (Chapter 5) — including a complete worked example followed end to end from a single reach toward a jar to a derived habit no sensor ever observed (Chapter 6). It then descends into the machine: the formal domain model (Chapter 7), the contract-first software architecture (Chapter 8), the salience-governed perception subsystem (Chapter 9), consolidation (Chapter 10), and grounded recall with calibrated refusal (Chapter 11). The remaining chapters treat the privacy architecture and its threat model (Chapter 12), the evaluation methodology that makes every failure diagnosable (Chapter 13), engineering governance (Chapter 14), the project's two stacks and the migration between them (Chapter 15), market position (Chapter 16), honest risks (Chapter 17), the strongest objections answered at full strength (Chapter 18), and the road ahead (Chapter 19).

The central hypothesis — that perception can be decomposed into channels which reconstruct lossless-enough textual context for a tractable reasoner to answer arbitrary retroactive questions at near-zero hallucination — is stated throughout as what it is: falsifiable and instrumented. Every architectural decision in this document exists to test it faster.

Executive summary

For the reader with ten minutes. Every claim below is developed, evidenced, and stress-tested in the body; section pointers are given throughout.

The product. TRACE is a wearable memory you can question. It senses the day through always-on, cheap perception; turns everything into words the instant it happens — never keeping one frame of video or one second of audio; binds those words into understanding at night; and answers any question about the past with citations to what it actually perceived, or an honest "I don't know."

The insight. The valuable artifact of a lived day was never the footage — it was the bound, queryable meaning. Storage-first wearables keep the toxic artifact (raw media: subpoenable, breachable, socially unacceptable) while deferring the valuable one (extracted, connected facts). TRACE extracts at the moment of capture and keeps only the value (Chapters 1–2).

The moat. Surveillance is not forbidden by policy; it is impossible by architecture. The only object that can leave the device is typed text carrying the identifiers of the observations it came from; everything else dies at a guard short enough to audit in one breath. The store is encrypted under the user's own key. A competitor cannot copy this posture without abandoning the stored-media foundation its product is built on (Chapters 12, 16).

The machine, in one breath. Cheap senses run continuously (detection, OCR, speech, sound events, motion); an attention scheduler spends a hard hourly budget of deep vision-model looks only where dwell, novelty, and stability earn them; every emission is a typed, timestamped, provenance-bearing claim in an append-only log; consolidation fuses claims by co-occurrence into living nodes — refusing every weak link — and derives new facts no sensor observed; at question time, a supplier hands a local reasoner just enough cited evidence to answer or refuse (Chapters 5, 9–11). Chapter 6 walks one memory through every stage.

The discipline. Seven locked invariants govern all code (Chapter 4): no raw persistence; text-only egress; append-only truth; provenance on everything; calibrated confidence with refusal over fabrication; unconstrained questions; and nothing built until a missed question demands it. Most are enforced by construction — an invalid memory object cannot exist.

The measurement. Two instruments keep the system honest (Chapter 13). RAS scores answers with fabrication subtracted — a made-up answer costs what a correct one earns. OAG measures what verbalization lost: of the questions a raw-footage oracle can answer, the share the retained text cannot. Every miss is mechanically diagnosed as (A) never captured or (B) captured but unused, and the miss list is the backlog. The first measured baseline was poor and is reported anyway (RAS 40.0, hallucination 27.3% on the proving clip); the refusal-calibration research line — run with research collaborator Latheesh Roy — subsequently drove confident-wrong answers to **zero on the reproducible benchmark battery**, and holding that zero on cold captures is a standing gate, not a claim of final victory (§13.5, §13.8). On the perception side the honest state is unsettled. Earlier drafts of this document reported 97.4% recovery against a "frontier-model oracle" over 111 real frames; on 2026-07-21 that figure could not be reproduced from any artifact in the evaluation tree and has been withdrawn. What the frozen artifacts actually hold is **83.9%** on the capability instrument (n=26) and **61.6%** on the completeness instrument (n=29, baseline frozen 2026-07-14), and the oracle in every one of them is `gemma3:12b-it-qat` — a local quantized 12B model, not a frontier one (§9.5).

The state of the build. The full thinking machinery — binder, consolidation, supplier, grounded recall — is implemented and passes an end-to-end crown test on synthetic weeks; the hardware senses are still adapters-on-fakes; and a v0 stack has run live capture daily since June 2026 while the greenfield rebuild strangles it organ by organ (Chapter 15). The ledger on the next page states exactly what runs, what runs on fakes, and what does not yet exist.

The hypothesis, stated as one. That textual channels can carry lossless-enough context for near-zero-hallucination recall is falsifiable and instrumented rather than assumed. The method: prove sufficiency offline with unlimited compute first; discover the necessary channel set by ablation second; optimize for the live budget last (Chapter 13, Appendix K).

What is real today

The reader deciding whether to join this project should not have to excavate the truth from an argument. This table is transcribed from the build ledgers of the two repositories on 2026-07-21. Each row names its evidence. Staleness warning: these are facts about a fast-moving codebase, correct on the date above and demoted to hearsay a month later.

Component	Status	Evidence
Domain model · claim log · rendering	IMPLEMENTED	`tracemem` M0–M1, green tests
Binder — reconcile, tombstones	IMPLEMENTED	`tracemem` M3
Consolidation — guardrails, selector, judge, pruner, remap	ON FAKES	`tracemem` M4 — implemented against synthetic inputs
Information supplier — cue, retrieve, pack, ground	IMPLEMENTED	`tracemem` M5
`.trace` container — format, keyring, sync, boundary	IMPLEMENTED / STUBBED	M6; libsodium crypto adapter stubbed
Extraction cascade	ON FAKES / STUBBED	M7 — logic real, hardware adapters (OCR/ASR/eyes/tap) stubbed
Crown test: shuffled synthetic week ⟶ episodes ⟶ entities ⟶ nightly patterns ⟶ cited answer ⟶ honest abstention	IMPLEMENTED	`tests/test_crown.py`, passing
Unit wall	IMPLEMENTED	274 green · 1 xfail (the TODO ledger, each naming its spec section)
Measured extraction score	WITHDRAWN	the 0.974/111-frame figure carried by earlier drafts could not be reproduced from any artifact on 2026-07-21 and is withdrawn. What the frozen artifacts hold: **83.9%** (capability, n=26) and **61.6%** (completeness, n=29, baseline frozen 2026-07-14), scored against a **local quantized oracle**, `gemma3:12b-it-qat` — not a frontier model. See §9.5
Confident-wrong answers on the benchmark battery	IMPLEMENTED / NOT YET	0% on the in-house battery — refusal-calibration line (with Latheesh Roy). Scored against a self-authored judge; **no cold-capture result exists**, so this is not yet an external claim
Falsification gate on real captured weeks	NOT YET	the gate needs real captured weeks — the next milestone that matters

Component	Status	Evidence
v0 stack: live capture, hub, nightly job, phone app	IMPLEMENTED	running since June 2026; being strangled, not extended

The number that matters most is one that is absent: frames kept — zero, by construction, in both stacks. There is no code path that writes a frame to disk.

How to read this document

This document is written for six kinds of reader. Do not read it linearly unless you want to.

- **In 10 minutes:** the Executive Summary, the ledger above, and Chapter 6.
- **The evaluator (an hour):** Chapter 1 (the claim) ⟶ Chapter 6 (the claim made concrete) ⟶ Chapter 13 (how failure is measured) ⟶ Chapter 17 (what could kill it) ⟶ Appendix C (the scorecard). This path contains every number and every risk; nothing in the remaining chapters softens or contradicts it.
- **The engineer (an afternoon):** Chapters 4–5 ⟶ Chapters 7–11 ⟶ Appendix F (the wire format) ⟶ then the repository itself.
- **The skeptic (an evening):** Chapter 2 (why words) ⟶ Chapter 3 (who tried before) ⟶ Chapter 18 (the objections, at full strength) ⟶ Appendix L (the case against, from the literature) ⟶ Chapter 12 (the moat and its honest residue) ⟶ Chapter 17 (the risk register).
- **Cognitive science:** Chapters 2, 6, and 10 — the biological arguments are load-bearing, and you are the reader best placed to break them.
- **Investors and diligence:** the ledger above, then Chapters 15–19.

Contents

Part I · Foundations

CHAPTER 1

Introduction and problem statement

1.1 The problem: lives are forgotten by the people who live them

People forget most of their lives. Where the keys were left. What the doctor actually said, as opposed to what anxiety later reconstructed. Whether the door was locked. The name attached to a face met once at a dinner. Which pharmacy the prescription went to. What the contractor promised, verbally, in the hallway. The moments that matter are sparse, unpredictable in advance, and irretrievable by the time they are needed. Human episodic memory was never designed for retrieval on demand; it is a reconstructive process, optimized for gist over fidelity, and it degrades precisely along the dimensions — exact words, exact places, exact times — that everyday questions require [1][2].

The obvious remedy is to record everything. It fails twice, and both failures are fundamental rather than incidental.

The first failure is social and legal. A person wearing an always-on camera that stores video is a walking surveillance device. Every bystander is recorded without consent; every stored hour is subject to subpoena, theft, breach, and misuse; every intimate space the wearer enters becomes an archive. The history of wearable cameras is a history of this failure: from the earliest sousveillance experiments to the public rejection of camera glasses, storage-first capture has been socially unacceptable everywhere it has been tried at scale [10][34]. No privacy policy fixes this, because policies are promises, and promises about stored media are exactly what nobody believes.

The second failure is functional. Raw footage is the wrong artifact. Nobody rewatches their life; a day of video is a day long. What people actually want from a memory is not playback but answers: they want to query the past the way they query a database, in natural language, and get back a short, true, cited response. The valuable artifact was never the footage — it was the bound, queryable meaning that a mind would have extracted from it. Storage-first systems defer that extraction forever and so never deliver it.

1.2 The problem statement

TRACE's problem statement follows directly:

> *Build a system that perceives a person's day through always-on, cheap senses; converts everything to words the instant it happens — never keeping one frame of video or one second of audio; consolidates those words, continuously and best nightly, into bound understanding; and later answers arbitrary questions from that understanding with citations, or refuses honestly.*

Three phrases in that statement carry the whole design, and each is defended at length in this document. "The instant it happens" is the privacy architecture (Chapter 12): ephemerality at the source, not deletion after the fact. "Bound understanding" is the binder and the node layer (Chapters 5, 10): isolated observations are raw material, not memory; memory is what emerges when the label, the spoken sentence, and the reach are fused into one moment. "Or refuses honestly" is the epistemology (Chapters 11 and 13): a memory that fabricates is worse than no memory at all, so refusal is a first-class outcome and fabrication is priced into the headline metric at the cost of a correct answer.

1.3 The central hypothesis

Everything rests on one hypothesis, repeated here because intellectual honesty about it is the whole game:

> *Perception can be decomposed into channels that reconstruct lossless-enough textual context for a tractable local reasoner to answer arbitrary retroactive questions at near-zero hallucination.*

It is falsifiable — Appendix K states it formally, and the oracle-answerability gap of Chapter 13 is its direct empirical estimator. It is instrumented rather than assumed. And it is not yet proven on real captured weeks; the falsification gate that would confirm or kill it is the project's named next milestone.

1.4 What TRACE is not

TRACE is not a lifelogger: it keeps no log of life, only text extracted from it. It is not a surveillance device: the artifact surveillance requires is never created. It is not a retrieval-augmented chatbot: retrieval here feeds a memory that was built for retrieval under a privacy invariant, not a pile of documents that happen to exist. It is not a model company: the language model is the last and least special box in the machine, ordinary and swappable by design. And it is not finished: the ledger in the front matter is the boundary between the built and the intended, and this document never blurs it.

1.5 Contributions of this document

First, a first-principles derivation of an input-only memory architecture from the biology of human memory and the failure history of lifelogging. Second, seven locked invariants, each with rationale, enforcement, and the failure mode it forecloses. Third, a complete pipeline exposition with one moment walked through every stage. Fourth, a formal domain model in which invalid memory is unrepresentable. Fifth, an evaluation method in which every miss mechanically names the work that fixes it. Sixth, a candid account of the project's two stacks and the strangling plan between them. Seventh, the strongest objections to all of the above, answered at full strength, including the counter-literature (Chapter 18, Appendix L).

CHAPTER 2

Conceptual foundations

2.1 The standard model: a brain never touches the world

The napkin's first drawing is not of the product; it is of every mind that has ever existed: `[World] → Eyes/Ears → [Brain]`, with the loop closing back into the world, annotated: *"because we can't manufacture other senses, or most of the information is obtained through this."* A brain never touches the world directly. It receives narrow, lossy sensory streams, extracts meaning at the moment of perception, and stores the meaning — never the stimulus. TRACE copies exactly this: video for eyes, audio for ears, extraction at the moment of capture, retention of meaning only. This is not biomimicry for its own sake; it is the observation that the only working memory system we know of is input-only, and that its constraints are load-bearing. Eyes and ears become camera and microphone; the napkin's "organs" become the sensor suite — pose, motion, place — the one sense a device can manufacture beyond sight and sound, and the one that answers spatial questions.

2.2 Episodic memory is reconstruction, not playback

Human recall does not replay footage, because there is none. Remembering is reconstruction from consolidated traces, schema-driven and confident even when wrong [2]. Tulving's episodic/semantic distinction [1] maps directly onto TRACE's two stores: the time-stamped claim log is episodic — events located in time and place, sparse and literal — while the bound evidence graph is semantic: consolidated, cross-referenced, queryable. The mapping is load-bearing, not decorative. It predicts the interface between the stores: consolidation runs episodic⟶semantic, never the reverse, exactly as the night shift reads the log and never writes it. And the system's honesty about reconstruction exceeds the biological original: where human recall silently invents schema-consistent detail, TRACE's recall is forbidden to assert anything that does not trace to a logged claim, and says "I don't know" where a human would confabulate.

2.3 Sleep consolidation: why the binder runs at night

The decision to concentrate deep binding in a nightly pass is likewise borrowed from biology. Systems-consolidation research shows that the mammalian brain replays the day's hippocampal traces during slow-wave sleep, gradually binding them into neocortical structure [3] — an architecture forced by the same constraint TRACE faces: deep integration is expensive, and the organism cannot afford it while also perceiving. Night is when the device is charging, thermally unconstrained, and free to run the big model over the whole day at leisure. The napkin's phrase is exact: node creation is *"always running in the background, but better nightly."* Incremental binding accelerates; the night shift perfects. The Hebbian slogan — cells that fire together, wire together [35] — is the binder's actual operating principle, transposed to text: claims that co-occur in time and place, repeatedly, get bound into shared structure.

2.4 Event segmentation: why capture is gated, not uniform

Human perception does not sample the world uniformly; it carves experience into events at boundaries of change, and encodes richly at those boundaries [4]. TRACE's capture follows the same economics (Chapter 9): the cheap channels run continuously precisely because they are cheap, while the expensive channel — deep visual description by a vision-language model — fires only on attention, dwell, or change. Uniform deep capture would be both unaffordable and pointless: most moments do not matter, and the value of a memory is concentrated in the few that do. The scheduler that makes this call is not an optimization detail;

it is the system's model of what deserves to be remembered well, which is why the project treats it as part of the moat.

2.5 Text as the retention format: the four arguments

The load-bearing decision — keep words, not media — rests on four arguments.

Privacy. Text is the only format whose retention can be made socially and legally safe by construction. A sentence about a bystander is a fundamentally different object from their stored face; the residue this does not discharge is treated honestly in §12.7.

Sufficiency. The hypothesis under test is precisely that words carry enough. Language is the interlingua in which modern machine reasoning is strongest [7], and retrieval-then-reason over text is the best-understood recall architecture available [12]. The strongest evidence against — the verbal-overshadowing literature, in which describing a face degrades later recognition of it [42] — is engaged directly in Chapter 18 and Appendix L, because it is a critique of human verbalization under a recognition criterion, not of machine extraction under a question-answering criterion, and the distinction is measurable.

Economy. Text is orders of magnitude smaller than the media it replaces — Appendix F's worked minute is roughly 2 KB against roughly 1.4 GB of 4K video for the same span — and a lifetime of it fits on a phone. "Never delete" (I3) is affordable only because of this ratio.

Auditability. A user can read every single thing the system knows about them, line by line — an impossible offer for embeddings or media, and the foundation of the trust the product sells.

2.6 The resolution of the central tension

"Remember everything" and "respect everyone" are in tension only while the retained artifact is media. Retain extracted meaning instead and the tension dissolves: recall quality becomes a function of extraction quality — which is measurable and improvable — while the surveillance artifact simply never exists.

CHAPTER 3

Prior art and position

3.1 The memex lineage: total capture as an old dream

Bush's 1945 memex [36] founded the genre and set its sixty-year trajectory: capture everything, organize later. The lineage's lesson is that "later" never comes — capture without retrieval-of-meaning is a landfill. Sellen and Whittaker's constructive critique of lifelogging [41] sharpened the lesson into design guidance a decade before this project: the value of a memory aid lies in retrieving the right cues and facts, not in total archives, and systems should be designed around the "five R's" of remembering rather than around storage. TRACE takes that critique as a requirement rather than an objection: it is a cues-and-facts system by construction, and the archive it refuses to keep is precisely the one the critique found worthless.

3.2 MyLifeBits and SenseCam: storage-first, and what it taught

Gemmell, Bell and Lueder proved the landfill empirically at life scale [9]: MyLifeBits' bottleneck was never storage but structure. SenseCam [10] proved the opposite pole — even passive photo capture measurably aids clinical memory — while every deployment surfaced the social cost of visible storage. Hoyle et al. quantified that cost from the wearer's own side: lifeloggers themselves censor, delete, and manage bystander exposure [34]. Gurrin, Smeaton and Doherty's survey of the field converges on cues-and-facts over archives [37]. Together these justify TRACE's inversion: total capture, linguistic retention.

3.3 Ego4D and egocentric vision: the demand curve, measured

Ego4D [11] defines the benchmark tasks TRACE's question batteries descend from — episodic memory queries against first-person video — and establishes that long-horizon egocentric retrieval is hard even with the video retained. That is the honest context for reading every recall number in this document: the ceiling is not 100%, even for an oracle holding the footage.

3.4 Retrieval-augmented generation: the supplier's ancestry

Lewis et al. [12] define the retrieve-then-generate shape the information supplier inherits. TRACE departs from RAG in what is retrieved from: not a corpus that happens to exist, but a memory constructed for retrieval under a privacy invariant, with provenance and confidence as type obligations rather than metadata conventions. Attribution — answers that cite — is promoted from an evaluation criterion [31] to a construction rule; selective prediction [13] and calibration [20] supply the formal frame for refusal; the hallucination taxonomy of Ji et al. [32] is operationalized as the grounding gate's checklist.

3.5 The commercial field: an empty quadrant

Two axes organize the field: what persists (raw media versus derived text) and where reasoning happens (cloud versus device). Screen-recording memory tools, audio pendants, and camera glasses all sit in the raw-persistence half-plane, differing only in which medium they hoard. The text-only, device-perception quadrant is empty, and not by accident: occupying it requires giving up the raw stream, which storage-first architectures cannot do retroactively — their features, their indexes, and their user expectations are built on the archive. Chapter 16 develops the consequence.

3.6 Knowledge graphs and open vocabulary

The node layer is a knowledge graph in the broad sense of Hogan et al. [14], with one deliberate deviation: open vocabulary everywhere. There is no fixed ontology of entity types or predicates; the world names itself through the extractors' text, and structure emerges from co-occurrence rather than from a schema. The cost of that choice is weaker global consistency; the payment comes at answer time, where evidence is quoted rather than inferred across long relational chains.

CHAPTER 4

The seven invariants

Architecture is what remains constant while everything else iterates. TRACE locks seven invariants; each is stated with its rationale, its enforcement mechanism, and the failure mode it forecloses. Code that violates an invariant does not merge — several are enforced by construction, so violating them does not even compile against the domain model.

4.1 I1 — No raw persistence

Statement. Raw pixels and audio are discarded the instant words are extracted from them. There is no record path, no cache, no “temporary” buffer that survives the perceptual moment.

Rationale. Every stored frame is a liability with no compensating asset: it cannot be queried directly, and it is the artifact whose existence makes the system a surveillance device. Ephemerality at the source is the only privacy claim that survives an adversarial audit — deletion policies, retention windows, and encryption of stored media all reduce to promises.

Enforcement. Capture code converts and drops in one motion; the repository’s ignore rules block media files as defense-in-depth; and in `tracemem` the law is inherited in its hardest form: *raw is immutable and no update path exists; pixels never leave the device.* Honesty note: evaluation gold frames exist on the founder’s development machine as oracles — they are the measuring instrument (§13.2), not the product path, and the phone build has never persisted a frame.

4.2 I2 — Text-only egress

Statement. The only payload type permitted to cross the device boundary is typed text carrying the identifiers of the source events it derives from.

Rationale. Local models may hit a quality wall (Chapter 17); the architecture must allow routing reasoning to a stronger model without ever routing media anywhere. Typing the boundary makes the safe path the only path.

Enforcement. In the v0 stack, an eleven-line egress guard raises a privacy violation for any payload that is not typed text with non-empty provenance — short enough to audit in one breath, which is the point. In `tracemem` the same contract lives in the container’s boundary module.

4.3 I3 — Append-only truth

Statement. The claim log is immutable and append-only; it is the single source of truth. Everything derived from it — the evidence graph, every index — is a disposable, replayable projection.

Rationale. Consolidation is experimental and will be wrong often; retrieval structures will be redesigned repeatedly. If derived structure were the truth, every such change would risk memory loss. With an immutable log, any projection can be thrown away and rebuilt — the napkin’s discarder rule: marinated content that proves useless is dropped, never the raw material.

Enforcement. The store exposes no update and no delete; corrections are expressed as new claims and tombstones (Chapter 7), never as mutation. Replays are deterministic, which Chapter 13′s evaluation depends on.

4.4 I4 — Provenance everywhere

Statement. Every claim, binding, citation, and egress payload carries the event identifiers it derives from. An unsourced assertion is a construction error.

Rationale. Citations are the product (answers must show their evidence), diagnosis is the method (a wrong answer must be traceable to the claim that misled it), and audit is the promise (the user can see where every remembered fact came from).

Enforcement. Constructors: a binding rejects an empty evidence tuple; a claim rejects a timestamp that disagrees with its provenance; an egress payload rejects empty source lists. Provenance is not a logging convention — it is a type obligation [19].

4.5 I5 — Calibrated confidence; refusal over fabrication

Statement. Every binding carries a confidence intended to be calibrated; links below the floor are refused, not hardened; at answer time, insufficient evidence yields "I don't know."

Rationale. A memory's worth is its trustworthiness. One confident fabrication — "you locked the door" when you did not — costs more trust than a hundred honest refusals. The metric agrees: RAS subtracts fabrications from corrects, so the optimizer and the ethics point the same way.

Enforcement. Confidence is range-checked at construction; the refusal floor is the binder's default disposition of weak evidence (Chapter 10); calibration itself is a measured property (Chapter 13), per [20]; and in `tracemem`, *grade degrades with derivation depth* is a hard law — deep derivations must be proportionally better-evidenced to be spoken.

4.6 I6 — Unconstrained questions

Statement. There is no fixed question taxonomy, no menu of supported query types. Users ask anything, natively.

Rationale. A constrained question surface is a product decision masquerading as a safety decision: it hides hallucination by refusing to hear the questions that would expose it. The honest path is open vocabulary end to end — open observation kinds, open predicates, open questions — with calibrated refusal as the safety mechanism instead of a menu.

Enforcement. The query type is a bare text string by design, and the founder's locked decision to keep it so is recorded in the decision log (Appendix D).

4.7 I7 — Nothing is built until a missed question demands it

Statement. Every helper, sensor, channel, and feature must be authorized by a diagnosed failure on a real question from a real capture.

Rationale. Perception systems invite infinite plausible work; intuition about which channel matters is reliably wrong. The project's own history supplies the cautionary tale: a 3D world renderer was built on intuition, answered nothing, and was killed. The discovery loop — ask, miss, diagnose, build exactly the missing channel — replaces guessing with evidence, and the overnight self-play harness automates the asking.

Enforcement. Governance (Chapter 14): the plan-gate requires naming the missed question before any build starts. The (A)/(B) diagnosis log supplies missed questions mechanically. Even the two `tracemem` modules that exceed the napkin — a prediction spine and the falsification harness — are flagged *beyond-napkin* in the constitution file, awaiting an explicit founder ruling: the invariant applied to the repository itself.

4.8 The invariants as a system

The seven interlock. I1 and I2 make the privacy claim structural; I3 and I4 make memory auditable and experimentation safe; I5 and I6 together make "ask anything" compatible with "near-zero hallucination" —

the refusal gate absorbs what open vocabulary exposes; and I7 aims the whole machine at evidence. Remove any one and the others weaken: without I3, diagnosis loses its replay; without I4, refusal loses its justification; without I7, the roadmap reverts to guessing.

Part II · The Machine

CHAPTER 5

The pipeline: from photons to answers

This chapter is the napkin's EXECUTION line, expanded stage by stage. Appendix A maps every napkin phrase to its production counterpart; the interactive companion (`interactive/metamorphosis.html`) renders the same pipeline as one continuous transformation. The pipeline in one line:

```
[World] → Extraction ■ → TEXT → Binder → Node Creation ■ → Supplier ■ → LLM
```

5.1 Stage 1 — Senses

Two streams open onto every second: video for eyes, audio for ears, plus the device's manufactured sense — pose, motion, place — standing in for the napkin's "organs." Neither audiovisual stream is a recording. Frames and samples exist transiently in working memory, are read, and die; what survives is only what was read from them. The cheap stratum (Chapter 9) watches continuously; nothing else is awake.

5.2 Stage 2 — Extraction

Extraction turns raw light and sound into addressed text, through four mechanisms. The *blur gate* triages every frame before anything else sees it: a frame whose content cannot be read — motion smear, defocus — is discarded immediately, and a frame that cannot be read is never written. The *spend dial* sets resolution and framerate by what the world is doing, not by a clock: a still room is sampled almost for free, a fast reach is frozen with a burst, a settled scene earns one expensive deep look. The *address stamp* fixes normalized coordinates and a timestamp on every reading before meaning is attached: nothing is remembered without its address, and any later claim can be walked back to the pixel and second that produced it. The *signal–noise decipherer* of the napkin is the composite of gate and dial: what remains after both is signal, stamped and contextualized.

5.3 Stage 3 — The macro picture

A vision-language model produces the napkin's *"macro picture of what's happening"*: a scene-level gist — a morning kitchen, someone reaching for a jar — that supplies context in both directions. Downward, it tells the scheduler what matters in this scene; upward, it rides with every emission so that later binding and answering know the setting a claim was born in.

5.4 Stage 4 — The helpers

With context established, specialists are called for the job: an OCR reader for the label, a state extractor for the jar, a detector for objects, ASR for the voice, a sound tagger for the room. Each is the best available understanding of its one slice of the world; each writes text; none knows the others exist. No helper is architecture — remove any one and the rest carry on — and the helper set is open-ended by design, discovered by ablation rather than capped by a guess (§13.4).

5.5 Stage 5 — TEXT

The output of extraction is the napkin's boxed word: TEXT. Text from all helpers, marked by coordinates, appended to the immutable log. Every line carries where, when, which helper, and how sure. This is the entire output of perception: text the owner could read aloud.

5.6 Stage 6 — The binder

The binder makes many readings into one moment. It does not understand the kitchen; it observes that several lines share a where and a when — *"binding the coordinates, physicality & temporal together"* — and fuses them on that evidence alone. Binding is evidence-driven, not semantic, which is why a wrong binding can be reconciled later without touching what was read. The output is the napkin's *first thought*: raw material that can be marinated on. The napkin's own note is the design's humility: *"this is not a complete Node, it's raw material."*

5.7 Stage 7 — Node creation

Raw material becomes living cells — *"a physical living breathing cell that can combine with fellow nodes to derive new things — 1+1 = 4, ≠ 2."* A node is a persistent thing (a jar, a person, a shelf) that thickens every time the world confirms it. Nodes combine: co-firing histories derive facts no sensor observed. The box carries the napkin's two governance rules: encryption — only the person using the app can decipher (Chapter 12) — and the discarder of meta: derived content that proves useless is deleted; the raw beneath it, never.

5.8 Stage 8 — The information supplier

A question does not go to the archive; it goes to the supplier, whose contract is the napkin's phrase *just enough context.* The supplier walks the graph, selects the few claims that bear on the question — three memories, not three thousand — and packs them with their citations for the reasoner. Retrieval quality, not model quality, is where answers are won, which is why the supplier is a named black box and the model is not.

5.9 Stage 9 — The LLM does what an LLM does

The last box is the least special. An ordinary local model [29] receives the packed context and writes the sentence, under a grounding gate that admits only clauses the packet supports. The model is swappable by design; the memory that feeds it is not.

5.10 What flows between stages

One direction, one format. World⟶Extraction: transient media, dead in milliseconds. Extraction⟶TEXT: addressed claims. TEXT⟶Binder: claims. Binder⟶Nodes: bound moments. Nodes⟶Supplier: evidence with provenance. Supplier⟶LLM: a small cited packet. LLM⟶user: a sentence whose every clause walks back to a claim, or a refusal. Nothing flows backward; nothing mutates upstream.

CHAPTER 6

One moment, end to end

The worked example is the napkin's own scene. Times are illustrative; the mechanics are the shipped ones, and Appendix F shows the same machinery as raw wire-format records from a real v0 capture.

6.1 09:14:07, Tuesday, the kitchen

A person reaches for a jar and says something: *"we're nearly out of the hazelnut stuff."* Light and sound. This is the only moment that will ever happen; everything after is about keeping it.

6.2 The same second, inside extraction

The blur gate discards the smeared frames of the reach and keeps one legible frame. The spend dial, idling on a still kitchen, spikes for the motion and settles into one deep look. The macro eye reads the scene — a morning kitchen, a reach toward a shelf — and calls the specialists. Three helpers write four lines:

```
09:14:07  x .44 y .48   label reads "NUSSGOLD"        verbatim 1.00
09:14:07  x .31 y .42   glass jar, ¼ full, lid ajar   gist 0.72
09:14:07  whole frame   morning kitchen, a reach      gist 0.61
09:14:09  mic           "…nearly out of the hazelnut  verbatim 0.93
                         stuff"
```

The background is transcribed too — a kettle click at 09:14:05, footsteps leaving at 09:14:12 — because one day the question will be "did I put the kettle on?", and the answer will exist. The frame that produced all of this lived roughly a hundred milliseconds. Frames kept: zero.

6.3 The binder, that evening

Same spatial region, same seconds: the label line, the jar-state line, and the spoken line fuse into one bound moment — *09:14, kitchen: the jar is nearly empty, and it was said aloud.* The chime from the neighbor's balcony, seven seconds and one wall away, is scored against the same arithmetic and refused (Appendix J shows the numbers). Raw material; not yet a node.

6.4 The night

The moment is absorbed into cells. The JAR node thickens — seen seven times since 12 May, a quarter full today. The PERSON node records that it was said aloud. The PANTRY shelf carries its forty-one sightings. Nodes fire together, and the night shift derives a third thing no sensor observed: *you run out roughly every six weeks* — graded as derived, citing the three readings that earned it. A second candidate, "prefers the 400 g size," finds no rent-paying evidence and is deleted. The raw beneath is untouched, so a better idea tomorrow gets the same evidence.

6.5 Weeks later: the question

"Do we need more of that spread?" The supplier walks the nodes and packs three memories: the quarter-full sighting with its date, the spoken sentence, the derived six-week cycle. The model writes: *"Almost certainly. It was a quarter full on 12 June, you said so yourself — and it's been about six weeks."* Each clause carries its citation. Ask instead "which brand does my sister buy?" and the answer is *"I have never observed that."*

6.6 What the example proves

That the pipeline composes: address-stamped extraction makes binding possible; binding makes nodes possible; nodes make derivation possible; provenance makes the final sentence auditable and the refusal principled. And what it does not prove: that extraction is rich enough on real, messy weeks — which is exactly what the falsification gate exists to test.

CHAPTER 7

The domain model

The vocabulary below is the live system's. The unit of memory is the *claim*; the v0 stack's equivalent object is the observation, and Appendix F shows its wire format.

7.1 Claim — the atom

A claim is one line of extracted text with its address and its provenance: what was read, where in the frame (normalized coordinates), when (timestamp), by which helper, and how it is graded. Claims are immutable and append-only (I3). The claim log renders back to a human-readable transcript — the memory can be read, in order, line by line. One field deserves special note, inherited from the v0 observation record: the *refutation cue*, in which a claim announces at birth what future evidence would disprove it. A shadow misread as a bird is expected to lose its track within a second; when the track instead persists and earns a deep look, the cue's condition fails and the claim's standing strengthens. Falsifiability is a field, not a philosophy.

7.2 Provenance — the birth certificate

Event identifier, source channel, capture time. Every structure above the log — episodes, entities, patterns, citations, egress payloads — carries the provenance of everything it was built from (I4). A wrong answer is therefore traceable, mechanically, to the claim that misled it.

7.3 Grade and confidence — the currency of trust

Confidence is a probability, range-checked at construction and intended calibrated [20]; grade orders trust across derivation depth. A claim read directly off the world — a verbatim label, a transcribed sentence — is witnessed; everything built above it carries a grade that degrades with derivation depth, a hard law of the repository. The refusal gate consumes grades: deep derivations must be proportionally better-evidenced to be spoken at answer time.

7.4 Episode — the bound moment

The binder's output: claims fused by co-occurrence in space and time into one event, carrying the evidence tuple that formed it. Episodes are projections — rebuildable from the log — and subject to *reconcile*: when later evidence shows a binding was wrong, a tombstone retires it and a corrected episode replaces it. Correction is new information, never mutation.

7.5 Entity and pattern — the living cells

An entity is a node with a history: the jar, the person, the shelf — each a list of dated confirmations, resolved across days by identity scoring (Appendix J.1). A pattern is a nightly derivation over entities: the six-week cycle, the morning routine. Patterns are the napkin's 1+1=4, and they are deliberately the most disposable layer in the system: the pruner deletes any that stop paying rent, and the raw claims beneath guarantee they can be re-derived or replaced.

7.6 Query, citation, and the answer contract

Answering is a typed pipeline: cue (read the question), retrieve (walk entities and episodes for bearing claims), pack (assemble just-enough context, every item citable), ground (emit only clauses the packet supports;

abstain otherwise). The contract's output is either a cited answer or an honest abstention; there is no third shape. The query type itself is a bare text string (I6).

7.7 Constructors, not validators

Throughout, invalid states are unrepresentable rather than checked-for: claims without provenance, confidences outside range, bindings without evidence do not pass construction [19][22]. A malformed memory is not a bug to find later — it is an object that never existed.

CHAPTER 8

Software architecture

8.1 The order of authority

The `tracemem` constitution file states a precedence unusual enough to quote: the napkin outranks the code; the code outranks the tests; the tests outrank the campaign documents; the campaign documents outrank the specs. Prose lost its authority in this project once already (Chapter 15), and the ordering makes sure the running system, not the latest document, wins arguments.

8.2 Contract-first modules

Every source file carries a contract header: what it OWNS, its IN⟶OUT, its TEST, and its spec reference. The module tree maps one-to-one onto the napkin's boxes — `extraction/`, `text/`, `binder/`, `nodes/`, `container/`, `supplier/`, `llm/` — so a reader can hold the drawing in one hand and the repository in the other. Implemented logic is tested green; unimplemented bodies raise `NotImplementedError`, and their acceptance oracles exist ahead of them as xfail-marked tests.

8.3 The test suite as ledger

The test run is the project's status report, by design: green is the regression wall (implemented and guaranteed); xfail is the TODO ledger, each entry naming the spec section it awaits; a failure is a real bug that outranks all other work. As of 2026-07-21: 274 green, 1 xfail, and the remaining stubs are exactly the hardware/model adapter set — the boundary between the thinking machinery (real) and the senses (fakes), stated plainly.

8.4 Hexagonal: ports and adapters

The v0 stack is hexagonal in the classical sense [21][22] — a pure domain, an application layer of use-cases, ports outward, adapters inward — and `tracemem` keeps the discipline where it matters most: every hardware and model dependency (OCR, ASR, camera eyes, the cipher, the local LLM) sits behind a port with a deterministic fake. The crown test exercises the entire pipeline on fakes; making an adapter real changes no domain code.

8.5 Event sourcing and disposable projections

The append-only log as truth and everything else as replayable projection is event sourcing [23] with CQRS-flavored read models [24], chosen for exactly the property Chapter 10 needs: consolidation can be wrong, thrown away, and re-run over history without loss. Replays are deterministic; evaluation depends on it.

8.6 The hard laws

Five laws inherited from v0′s operational history bind all of it: raw is immutable — no update path exists; pixels never leave the device; one model on the GPU at a time (a thermal and correctness constraint learned when live capture went dark, silently, under a shared GPU); grade degrades with derivation depth; and every capture gap is witnessed — if the senses were dark, the memory must say so rather than paper over it.

8.7 The crown test

One test stands above the suite: a shuffled synthetic week is fed through the whole machine — episodes form, entities accrete, nightly patterns derive, a question is asked, a cited answer comes back, and an unanswerable question is honestly abstained from. It passes. It is the executable form of Chapters 5–7, and it is what “the thinking machinery exists” means when this document says it.

CHAPTER 9

The perception subsystem

9.1 The cheap stratum: always-on helpers

Perception is two economies. The cheap stratum — detection [26], OCR, speech recognition [30], sound-event tagging against the AudioSet ontology [39], motion and pose [15] — runs continuously because it can: these models are small, neural-engine-friendly, and power-light. Open-vocabulary naming rides on MobileCLIP [27], chosen by audit for exactly the properties the substrate demands: fast on the neural engine, words rather than prose, vocabulary growth without retraining. The expensive eye — deep description by a vision-language model [17][18][38] — runs rarely, because it must.

9.2 Salience: deciding what deserves the expensive eye

The scheduler prices every track by dwell, novelty, and stability — a wearable-economics translation of the computational-attention literature [8]. A still room earns almost nothing; an entering hand spikes the framerate; a settled new object earns the deep look. This is event segmentation (Chapter 2) turned into a scheduler.

9.3 The budget: a token bucket with no overdraft

Deep looks are governed by a token bucket [28]: a hard hourly budget, refilled at a measured rate, spent by salience, with no overdraft path. Attention is a budget, not a vibe; the system can always account for why it looked hard at one moment and not another, and the budget's refill rate is an empirical property of the device, not a config guess.

9.4 The progressive enrichment ladder

Attention deepens in rungs — glance (a jar), read (the label), state (three-quarters empty, lid ajar), context (a German promo sleeve, and therefore probably a German kitchen) — and each rung emits only the delta: what the previous rung did not already know. The same fact is never stored twice; storage grows with new knowledge, not with time spent looking.

9.5 The measured frontier

A correction, stated first because it is the kind of error that gets believed. Earlier drafts of this chapter reported L1-verbatim recall of **0.974** over **111 real captured frames** scored against **pinned frontier-model ceilings**. Re-derived against the repository on 2026-07-21, none of those three claims survives:

- **The oracle is not frontier.** Every oracle declaration in the evaluation tree — all four — reads `gemma3:12b-it-qat`: a locally hosted, quantized 12B model. No frontier model was ever called. A score of 97.4% against a 12B local reader is a materially weaker statement than the same number against the best reader available, and it is the only one the artifacts support.
- **The corpus is smaller than reported.** `evaluation/held/oracle_cache/` holds **29 frames carrying 872 claims**, not 111 frames carrying 4,674. The graded runs report `n=26` and `n_graded=29`.
- **The score is not reproducible.** No artifact yields 0.974. The scores that exist are **83.9%** (`extraction_capability.json`, n=26) and **61.6%** (`extraction_completeness.json`, n=29) — the latter also frozen as the baseline in `extraction_baseline_manifest.json` on 2026-07-14.

The two surviving instruments measure different things and neither is a drop-in replacement for the withdrawn figure, so no substitute headline is offered here. The per-class breakdown is the more useful object anyway, and it is unflattering exactly where it matters most: on the completeness instrument the `number` class retains 62 of 109 graded claims (17 corrupted, 30 missing) and `imagery` retains 32 of 96. Numbers and images are what a memory is asked to return; they are the weakest classes in the stack.

What does survive from the original audit is the diagnosed lesson — a cropping law confirmed on meal scenes (2/4 recovered on crops versus 0/4 on full frames) — the miss-diagnose-fix loop of I7 operating on perception itself. The standing caveats also survive and are worth restating: the mechanical tier of the claim-match instrument is not the full score, and frames are frames, not weeks.

CHAPTER 10

Consolidation: the night shift

10.1 The job

Read the day's claim log; bind by co-occurrence; thicken entities; derive patterns; refuse weak links; discard failed meta; touch nothing raw. The napkin's underlined words — *always running in the background, but better nightly* — set the schedule: incremental passes keep the memory current; the night pass, on a charging and thermally free device, does the deep work. Congregate, dedupe, bind; resolve identities (today's jar is the same jar, thickened, not a fourth jar); run the derivation pass; rebuild the projections; toward dawn, let the discarder drop what the night's better judgment replaced.

10.2 The anatomy

The consolidation heart is built as guarded stages: *guardrails* (classes of assertion that may never be derived, because they would outrun any evidence); four *derivation kinds* (the legitimate shapes of 1+1); a *selector* (which co-firing histories deserve the model's attention tonight); a *judge* (does the candidate cite enough to survive — refusal is the default disposition); the *night* runner; the *pruner* (rent collection on old patterns); and *remap* (re-pointing structure when reconcile retires an episode). All of it runs today against synthetic weeks; its inputs become real when the extraction adapters do.

10.3 Accelerator, not gatekeeper

Consolidation makes answers better and faster; it must never become the condition for answering at all. The supplier can serve from episodes alone — unconsolidated memory is slower and flatter, not absent. Chapter 6′s pharmacy-bag class of question — asked the same evening, before any night shift — is answered from the raw scaffold with a single citation. A memory that only works after a good night's sleep is a demo, not an organ.

10.4 The devil's advocate, kept in the room

Derivation is where a memory system starts lying to itself, which is why the judge prices skepticism in: every candidate pattern must survive an explicit attempt to refute it from the same evidence, weak links are refused at write time rather than hardened, and the system deletes its own failed thoughts without sentiment. The napkin's discarder is not cleanup; it is epistemic hygiene.

CHAPTER 11

Recall: grounded answering with calibrated refusal

11.1 The loop

Cue ⟶ retrieve ⟶ pack ⟶ ground. The question is read for its entities, time hints, and kind; the supplier walks the graph; the packet is assembled small — just enough context, every item citable; and the grounding gate lets a clause through only if the packet supports it. The reasoner is local [29] and ordinary; on v0 the heuristic fallback refuses everything, which is the correct degenerate behavior for a memory: no reasoner, no guessing.

11.2 The grounding gate

The gate's checklist is the hallucination taxonomy [32] operationalized, inherited clause by clause from the legacy engine's scar tissue (Appendix E of the strangling plan): binary-state discipline — occupancy, open/closed, on/off asserted only on direct literal support; phantom-class defenses on existence questions, where prior plausibility most strongly tempts invention; absence and negation handled from positive contrary cues, the hardest honest answer for a generative system; and over-specificity suppressed — the answer may not be more precise than its evidence.

11.3 Refusal as calibration

Refusal is not a failure mode; it is the mechanism that makes open questions (I6) safe. The formal frame is selective prediction [13]: answer only above a confidence bar, and measure the risk–coverage trade rather than assert it. Attribution completes it [31]: citations are checkable objects, so a wrong answer is a diagnosable event. The refusal-calibration line (§13.8) is where this chapter's theory earned its first zero.

11.4 What answering feels like

Short, cited, and honest. *"Almost certainly — a quarter full on 12 June, you said so aloud, and it's been about six weeks."* Or: *"I have never observed that."* The product bet, stated in Chapter 18.12: users forgive a memory that sometimes says "I don't know" and never forgive one that lies once.

Part III · Reality

CHAPTER 12

The privacy architecture

12.1 Layer 1 — Ephemerality at the source

Raw media is converted and dropped in one motion (I1). There is no retention window to configure, no deletion policy to trust, no archive to breach. The privacy property is not that stored media is protected; it is that stored media does not exist.

12.2 Layer 2 — The typed boundary

Only typed text with provenance may leave the device (I2). The guard is deliberately tiny — eleven lines in the v0 stack — because auditability of the boundary is itself a security property.

12.3 Layer 3 — Encryption under the user's key

The text memory is encrypted with AES-256-GCM [33]: key derived from the user's secret, a random 96-bit nonce per payload, and the container format version bound as associated data, so ciphertexts authenticate their own container. The napkin's phrase — *"encryption, i.e. people using our app can decipher"* — is this codec: unreadable to the platform, the cloud, and the thief; readable only to the user's own application holding the user's key. The `.trace` container carries three layers with deliberately asymmetric guarantees: a header readable without the key (version, device, counts); the RAW claim log, append-only and encrypted; and the META layer — nodes, edges, confidence — encrypted, deletable, and re-derivable. Lose the meta and nothing is lost; lose the raw and everything is. Status honesty: `tracemem`'s cipher adapter (libsodium) is presently stubbed, and the ledger in the front matter says so.

12.4 Layer 4 — Repository and process hygiene

Defense-in-depth extends to the development process: the repository's ignore rules block raw video, audio, and still formats globally; the commit gate verifies zero media binaries staged; the CI gate runs on every push. The team's own workflow is subject to the invariant it ships.

12.5 The threat model, tabulated

Threat	Storage-first system	TRACE
Device theft	media archive exposed	ciphertext under user key; text-only plaintext
Cloud breach	media archive exposed	nothing off-device but guarded text with provenance
Subpoena	stored footage discoverable	no footage exists; the text store is the complete universe
Insider access (vendor)	policy-limited	architecturally empty: no media to access
Bystander recording	permanent images of non-consenting parties	no image persists; bystanders appear only as typed gist

Threat	Storage-first system	TRACE
Function creep (a future feature "re-opens" media)	one flag away	impossible: the media was never kept

Appendix I walks four of these as scenarios, including the hardest one — the coerced unlock — where the honest answer is that text is real exposure, bounded to what words carry.

12.6 Regulatory alignment

Text-only, minimal, purpose-bound retention is data minimisation under GDPR Article 5(1)(c) [6]; capture-time conversion is privacy-by-design in the original, architectural sense [5]; user-key-only decryption aligns with data-protection-by-default. None of this is claimed as legal clearance — wearables face jurisdiction-specific recording law — but the architecture starts from the strongest position a perception device can occupy: the sensitive artifact class is never instantiated.

12.7 What architecture cannot discharge

Honesty requires the residue stated. Perceiving people into text still touches privacy: "Anna said she's pregnant" is sensitive with no pixel involved. The bystander literature on wearable cameras [34] transfers in part to any wearable perception. Product-level obligations therefore remain: disclosure norms for wearers, redaction policies for bystander speech, retention and export controls in the user's hands, and the wearer's own social contract. The architecture makes the worst artifact impossible; it does not make the remaining ones weightless. This section exists so that no investor, customer, or regulator can say the project hid the residue behind the moat.

CHAPTER 13

Evaluation: every miss names the next build

Measurement is the most opinionated subsystem in TRACE. It is designed so that failure is never ambient — every miss is attributed, mechanically, to a cause that names the work that fixes it.

13.1 RAS: fabrication priced like a loss, because it is one

The headline metric over a blind adversarial battery on a real capture:

$$\text{RAS} = (\text{correct} - \text{made-up}) / \text{total}$$

A fabricated answer subtracts what a correct one adds; an honest refusal costs nothing. The metric encodes the product ethics (I5) so directly that optimizing the number and behaving honestly are the same act. The qualitative bar attached to it: a stranger asks about your day and is amazed, with near-zero made-up answers.

13.2 OAG: the price of throwing the pixels away

The oracle-answerability gap: among questions an offline oracle with the raw frames can answer, the percentage the retained text cannot. OAG is the direct measurement of the retention decision's cost (§2.5) and the empirical estimator of the central hypothesis (Appendix K). Raw media's only legitimate role in the system is here — oracle-side, offline, never entering production recall. In `tracemem` the same idea is systematized as pinned *ceilings*: cached, versioned readings of the same frames, so every extraction score is a fraction of a known maximum rather than a free-floating number. Current audited state, re-derived 2026-07-21: **29 ceiling frames carrying 872 claims**, read by `gemma3:12b-it-qat` — a local quantized model. Earlier drafts described these ceilings as "frontier-model readings" over 111 frames and 4,674 claims; that description was wrong on all three counts and is withdrawn (§9.5).

This matters beyond the arithmetic. A ceiling set by a 12B local model is not a ceiling on what is **readable** — it is a ceiling on what **this reader** read, and the extraction stack can score highly against it while both miss the same things. The design intent stands: ceilings should be re-pinned against the best reader rentable, because that is the only honest definition of "lossless-enough." That pinning has not yet been done, and until it is, every extraction score in this document is relative to a local model rather than to the frontier.

13.3 The (A)/(B) instrument

Every miss is classified: (A) the fact was never captured — the context was not lossless enough — or (B) the fact was in the store and the brain failed to reach, hold, or reason over it. The classification is mechanical, not judged: every answer logs what the reasoner looked at, so a miss either had the evidence in view or did not. (A) misses authorize new channels (I7); (B) misses direct work to binding, retrieval, or refusal calibration. If a miss cannot be classified, the instrument itself is declared broken.

13.4 The method: sufficiency, then necessity, then efficiency

Three questions, answered strictly in order, never conflated. *Sufficiency* (brute force, offline): on a recorded clip, thermal and token limits do not apply; throw everything — deep VLM on every frame, full pose track, OCR everywhere, full audio — and ask only whether the brain can answer the hard set at approximately zero hallucination at all. This isolates "does the idea work" from "can we afford it." *Necessity* (ablation): remove one channel at a time and re-score; a removal that breaks no answer is not load-bearing, and the helper set is thereby discovered rather than guessed. *Efficiency* (last): only with the necessary set known does the salience scheduler return, to approximate that set within the live budget. Optimizing before sufficiency is proven is how projects die with beautiful, useless engineering.

13.5 The first honest baseline

The v0 proving ground was a 92-second outdoor walk with a 25-question adversarial set, scored from frozen answers: 16 correct, 6 wrong, 3 refused — RAS 40.0, hallucination 27.3%. The diagnosis: 8 of 9 misses were (B) — wrong-object bindings, confident-wrong binary states, one counting failure, retrieval misses — and 1 was (A), the head-pose channel that two spatial questions demanded, correctly refused rather than fabricated. The number was poor, the diagnosis was precise, and both are reported because a document that hides its worst number teaches readers to distrust its best one. Appendix C carries the full scorecard.

13.6 Anti-overfit machinery

Gold answer keys are hash-pinned; a silent edit to an answer key breaks the build. Datasets carry tuning-exposure flags, and the baseline's own validation clip is flagged as tuning-contaminated rather than quietly promoted. The pitch-readiness bar includes a cold second capture — never tuned on — passing a private regression check before any external claim. Question batteries are written blind: the author has not seen what the system captured, so the battery samples the world's distribution, not the system's strengths.

13.7 The self-play flywheel

Question generation itself is automated: an overnight job drives batteries against the day's memory, mines the misses, and files them into the (A)/(B) queue. The system that answers questions by night also discovers, by night, which questions it cannot answer — turning I7′s discipline into a continuously running loop rather than a quarterly ritual.

13.8 From 27.3% to zero: the refusal-calibration line

The gap between the baseline and the invariant was the project's daily work for weeks, and it is the workstream where the second contributor deserves naming: the refusal-calibration and benchmark line, run with research collaborator **Latheesh Roy** under the founder's direction, tightened the grounding gate and the confidence floors until confident-wrong answers reached **zero on the reproducible benchmark battery** — the measured basis for the product claim "it cannot lie to you," demonstrated as: it runs, it cites, it refuses. Two honesty notes bound the result. The zero is a battery property, not a theorem: it holds on the frozen adversarial set and its planted unanswerables, and the cold-capture guard (§13.6) exists precisely because a zero that has not survived a never-tuned capture is a provisional zero. And the zero is bought partly with refusals: driving made-up answers to nothing raises the refusal rate, and whether users accept that trade is a product question (§18.12) the beta exists to answer.

CHAPTER 14

Engineering governance

A small team with an ambitious architecture survives on discipline that is cheaper to follow than to break. TRACE's governance is code where possible, ritual where necessary.

14.1 The merge gate

One command enforced identically three ways — locally, as the pre-push hook, and in CI on every push: formatting and linting; strict typing over domain and application; complexity and function-length ceilings; dead-code detection with zero tolerance in production code; frozen-gold hash verification; the focused evaluator tests; and a coverage floor over the domain (87% at the v0 baseline). The gate is the definition of done.

14.2 One source of truth, read first, updated last

The repository's constitution and specs govern execution. Every working session starts by reading them and ends by updating them. The rule exists because the project's early history includes divergence between documentation and reality — including an acceptance failure traced to desk-testing a stale build — and the fixes are procedural: a build stamp (the git SHA in the app's status line and in every posted artifact), artifact-level verification, and the standing instruction to never trust "installed."

14.3 The plan-gate and the anti-diversion law

Before any build: name the roadmap item and the missed question that authorizes it (I7). No authorization, no build. The law's origin is a real casualty — the 3D world renderer, built on intuition, needed by no question, killed on diagnosis — and its function is to make that class of detour structurally difficult. Corollaries: pruning is measured, never aesthetic; and re-evaluation of locked component choices requires new diagnostic evidence, not new enthusiasm.

14.4 Honesty as an artifact

Phase reports record what was verified against the live code, including corrections of prior claims; evaluation reports carry their contamination flags; the README states which invariants are proven and which are targets. The document you are reading inherits the same rule — see §4.1, §9.5, §12.3, §13.8 — and the front-matter ledger is its enforcement.

CHAPTER 15

Two stacks, one product: an honest history

15.1 v0: the stack that runs

The reader doing diligence will find two repositories, and deserves the story before the surprise. v0 has run continuously since June 2026: live capture feeding an always-on hub, a nightly consolidation job, an encrypted store, an iPhone capture app, and the evaluation battery. It produced the first disciplined binder — 36 entities, 58 bindings, 144 weak links refused on the proving clip — the first honest scores (§13.5), the refusal-calibration zero (§13.8), and most of the hard laws (§8.6). It also accumulated what every first stack accumulates: prose documents that drifted from the code until, in mid-July 2026, the founder deleted every one of them and declared that the present tense lives only in code, tests, and running processes. That deletion is why this document cites ledgers instead of promises.

15.2 The greenfield: the napkin, rebuilt

Days later the architecture was redrawn by hand — the napkin — and `tracemem` was scaffolded from it: specs first, contract headers in every file, fakes behind every port, the crown test as the spine, xfail as the TODO ledger. The napkin was transcribed verbatim into the repository as its constitution, outranking everything including the specs. v0′s lessons are inherited as laws rather than as code.

15.3 The legacy engine: what v0 knows that a rewrite would forget

v0′s answer engine — roughly 2,900 lines that produced every baseline number — predates the domain model and contains 47 broad exception handlers. It is also a repository of earned answer-quality defenses, bought one hallucination at a time on real captures, and the strangling plan's first rule is that none of them may be lost in translation: the binary-state discipline (a sound event does not prove a crowd; a stray cable does not prove a phone was connected); the phantom-class watchlist guarding existence questions; absence-and-negation machinery for answering "no" from positive contrary cues; cross-lingual matching, learned from captures in a bilingual environment, where English questions must match German OCR; and the dossier-building craft that Chapter 11 formalizes as the supplier.

15.4 The strangling sequence

The migration follows the strangler fig [25][40]: v0 keeps running as the capture organ and the measuring baseline while `tracemem`'s adapters come real in dependency order — cipher, OCR/ASR, the camera eyes, the phone tap. Every extracted capability lands behind a port with tests that encode the legacy behavior it must preserve, and the baseline battery re-runs after each extraction. The operational scar tissue transfers with it: capture dies silently under a shared GPU and must be watched by liveness checks; embeddings needed a different engine than the reasoner; phone deploys stall in reproducible ways; a store must be snapshotted and shadow-served to be measured without killing capture. v0 is decommissioned by explicit founder decision when the falsification gate runs green on the new stack — not by drift, and not before.

CHAPTER 16

Market position and the moat

16.1 The quadrant

Two axes organize the field: what persists (raw media versus derived text) and where reasoning happens (cloud versus device). Screen-recording memory tools, audio pendants, and camera glasses all sit in the raw-persistence half-plane — differing only in which medium they hoard. The text-only, device-perception quadrant is empty, and not by accident: occupying it requires giving up the raw stream, which storage-first architectures cannot do retroactively — their features, their indexes, and their user expectations are built on the archive. An incumbent cannot follow without abandoning its own foundation. That is what makes the position a moat rather than a feature.

16.2 Why now

Three curves cross. On-device AI crossed the capability threshold: billions of phones run local models, and sub-second on-device vision is commodity. AI wearables ship in volume — roughly seven million camera glasses in 2025 alone — proving demand for the form factor while their retention architecture proves the objection: the always-on recorder category is a graveyard, its best-funded entrant shut down with assets sold to HP for $116M in February 2025. And the EU AI Act's biometric provisions, in force since February 2025, turn "structurally non-retaining" from a philosophy into a procurement criterion, on the continent where this company is being built.

16.3 Why incumbents want it and cannot build it

For platform acquirers, an input-only memory is the wearable strategy without the surveillance liability that has repeatedly burned camera products. Their business models monetize retained data; their privacy stacks deliberately stop at the digital life. The acquisition logic is strengthened, not weakened, by the moat's nature: the asset is an architecture and its measured evidence — the invariants, the evaluation trail, the scheduler — none of which can be bolted onto a storage-first product without a rewrite indistinguishable from starting over. If the capability matters to them, they license it or acquire it; both outcomes require TRACE to exist first. The licensing precedent is the layer companies — Arm and Dolby own a layer, not a box — and the `.trace` container plus the text-only context boundary are designed to be exactly such a licensable layer.

16.4 The demo that proves the moat

The pitch deliverable is designed to make the architecture felt: a hero capture answered live with citations; the no-raw-media proof panel — frames visibly converting to text and dying, the store shown text-only; and the live "ask it anything" moment where an honest refusal lands as a feature. An audience that watches surveillance become structurally impossible on stage has understood the company.

CHAPTER 17

Risks and open problems

Stated plainly, ordered by severity, with the mitigation that exists and the trigger that would escalate each.

17.1 The central hypothesis may be false

Text may not carry enough of the visual world. Mitigation: OAG and the ceilings exist to measure exactly this; the sufficiency phase brute-forces the best case before efficiency constrains it; the measured frame-level gap is withdrawn and currently unmeasured against a frontier ceiling (§9.5). Escalation trigger: a sufficiency run on a clean real week in which the brute-force text scaffold still misses a material share of oracle-answerable questions. That result would demand rethinking the retention format itself, and no amount of scheduling cleverness would matter. The project fails honestly in that world rather than pivoting to stored media.

17.2 The zero must survive the cold

Confident-wrong answers are at zero on the reproducible battery (§13.8); the risk is distribution shift. Mitigation: the cold-capture guard — a never-tuned capture must pass a private regression before any external claim; hash-pinned golds; contamination flags. Escalation trigger: hallucination reappearing on cold captures under a disciplined binder would point at the reasoner, opening the measured text-only hatch (I2-safe by construction).

17.3 The senses are not real yet

The thinking machinery runs; the hardware adapters are stubs. The risk is not that the adapters are hard — v0 proves each exists — but that their quality on device, on battery, all day, is below what the ceilings assume. The battery and thermal budget on glasses-class hardware is unmeasured, and this document says so rather than extrapolating.

17.4 Local models may be the wall

The local reasoner and VLM [17][18] may cap binding or answering quality. Mitigation: the hatch — binding and answering can escalate to a frontier text-only model with the privacy posture unchanged, because text was the only thing allowed out from the first commit. The decision is a benchmark delta, not a philosophy. Residual: dependence on a frontier vendor for the deep pass, bounded by the fact that only guarded text ever leaves.

17.5 Consolidation may not hold at life scale

If disciplined binding degrades over weeks — entity resolution drifting, derived nodes accumulating subtle wrongness — the graph's value erodes. Mitigation: accelerator-not-gatekeeper (the scaffold always answers); replayability (every binder version can re-run history, I3); refusal (the graph prefers silence to error). Honesty: this is the research risk. The crown test binds cleanly; a life is noisier than a test.

17.6 Two stacks is a tax, and one founder built this

Every week both stacks run is a week of split attention; the strangling plan bounds it, and the tax is admitted. Bus factor is one, and the seat this document exists to fill is empty (Chapter 19). The mitigations are the ones in evidence: a constitution, contracts, tests as ledger — a repository built to be joined — plus the collaboration already running (the refusal-calibration line, §13.8).

17.7 Bystander and social risk

Architecture removes stored media; it does not remove the social fact of being perceived. §12.7′s product obligations stand; recording-law heterogeneity across jurisdictions is a compliance program, not an architecture patch; and the possibility that society rejects even text-only wearable perception is priced in Chapter 18.3 rather than assumed away.

CHAPTER 18

Objections and responses

Every serious reader of this architecture raises a version of the same dozen objections. They deserve direct answers, in one place, at full strength. Where an objection is partially right, the response says so.

18.1 "If you throw away the pixels, you'll throw away the answer."

The strongest objection, and the one the whole evaluation apparatus exists to face. It is a measurable claim, not a debate: OAG quantifies exactly the questions the raw-frame oracle answers that the text cannot; the sufficiency phase maximizes the text side before any efficiency constraint bites; and the enrichment ladder exists to spend deep perception where tomorrow's question is likeliest to land. The honest current state: the frame-level gap against a frontier ceiling is **unmeasured** — the figure earlier drafts cited here was withdrawn on 2026-07-21 (§9.5) — the week-level gap is likewise unmeasured, and if the gap ever proves irreducible at the sufficiency limit, the hypothesis is falsified and the premise fails — a possibility the risk register states rather than hides. What the objection cannot claim is that the alternative escapes the problem: storage-first systems defer extraction, they do not solve it, and they pay the surveillance price forever while deferring.

18.2 "A verbal description of a scene is a pale shadow of the scene."

True — and the wrong comparison. The competitor is not the scene; it is human memory of the scene, which is itself a sparse verbal-categorical trace reconstructed on demand [1][2]. TRACE does not need to beat the videotape; it needs to beat the wearer's own recollection, with citations, at near-zero fabrication. The strongest scientific form of this objection is the verbal-overshadowing result [42]: describing a face in words degrades later recognition of that face. Appendix L takes it seriously, and the two-part response is structural. First, overshadowing is a human phenomenon — the verbal trace interferes with the human's own perceptual memory; TRACE's extraction does not overwrite anyone's memory, it supplements it with a second, citable one. Second, the criterion differs: overshadowing hurts recognition (pick the face from a lineup), while TRACE is evaluated on question-answering over facts — and where a described detail is genuinely unrecoverable from text, OAG counts it, which is exactly what the instrument is for.

18.3 "People will not wear a camera, period."

The social objection. People already carry always-on microphones and cameras in their pockets and on their wrists; what they reject — and what jurisdictions legislate against — is retention and replay. TRACE's answer is not "trust us" but "there is nothing to distrust": no stored image of any bystander exists one second after the moment. That claim is demonstrable on stage and auditable in code. It may still fail socially — §12.7 and §17.7 keep that residue on the books — but it fails from a categorically stronger position than any recording device.

18.4 "Local models are too weak to bind a day correctly."

Possibly. The architecture's response is the measured hatch: binding and answering can escalate to a frontier text-only model with the privacy posture unchanged (I2). The decision is a benchmark delta, not a philosophy: local is preferred, cloud-text is permitted, raw egress is impossible. The objection's teeth are economic, and the mitigation is the same as everywhere in this design: spend depth only where salience earns it.

18.5 "The graph will silt up with garbage over months."

The scaling objection, and the project's honest research risk (§17.5). Three structural answers: weak-link refusal keeps coincidence out of the graph at write time; projections are disposable, so a better binder can re-marinate history at any time (I3); and the graph is an accelerator, never a gatekeeper — recall can always fall back to the raw scaffold. What these do not answer is whether any binder discipline holds at life scale; that is what the phased evaluation is for.

18.6 "Why not just embeddings? Vector search over frames is simpler."

Embeddings are in the system — as projections, rebuildable accelerators for retrieval. They are disqualified from being the truth for three reasons: they are not auditable by the owner (no one can read what a vector of their kitchen contains); they are not citable at the claim level (I4); and they freeze a model's worldview into the store — a new encoder means re-embedding, which is fine for an index and catastrophic for a memory. Text is the only format that is simultaneously queryable, auditable, citable, and model-independent.

18.7 "OCR and ASR errors will poison the memory."

They will enter it — every channel is noisy. The defenses are layered: per-claim confidence at capture; cross-channel corroboration at binding (an OCR misread that nothing else co-occurs with binds to nothing); the refusal floor at answer time; and the refutation-cue field, which invites later evidence to overturn earlier error. The design position: noise in an auditable text store is a manageable disease; noise in an unauditable one is undiagnosable.

18.8 "This is just RAG with extra steps."

The recall stage is RAG-shaped, and the document says so [12]. The extra steps are the product: a corpus that is machine-perceived under a privacy invariant rather than scraped; provenance as a type obligation rather than a convention; refusal as a scored, calibrated outcome rather than a failure; and a nightly consolidation layer that RAG systems do not have because their corpora do not accrete a life. Dismissing a system by naming its weakest structural analogy is a critique this chapter welcomes — it is how the 3D renderer died — but here the analogy covers one of nine stages.

18.9 "The salience scheduler will miss the moment that matters."

Sometimes it will — a budget is a bet. Two answers. First, the cheap stratum never blinks: OCR, ASR, sounds, and sensors record continuously, so a missed deep look is a degraded memory, not an absent one. Second, every such miss is visible: it surfaces as an (A)-diagnosed failure, and the budget, weights, or ladder thresholds move in response (I7). The scheduler is not claimed optimal; it is claimed instrumented — its errors are the training signal for its own next version.

18.10 "Encryption with no vendor key means no recovery."

Correct, and chosen. A memory this intimate with a vendor-side recovery path is a memory with a second reader. The design accepts the consumer-grade consequence — lose the key hierarchy, lose the memory — in exchange for the categorical claim of §12.3; key-escrow-by-user-choice can exist as product surface without weakening the default.

18.11 "A team this small cannot build this."

The scope objection. The architecture is its own answer: a domain layer small enough to read in a sitting; one substrate at a time behind ports; a legacy engine strangled rather than rewritten; a machine that generates its own backlog from missed questions and runs its own night shifts. The governance chapter is not process theater — it is how a small team rents the discipline of a larger one. The risk that remains is focus, and the anti-diversion law exists because the team has already paid once for losing it.

18.12 "Why would anyone pay for honest refusals?"

Because the alternative is confident fabrication about their own life, and one such fabrication ends the relationship with the product. Refusal is not the product; trust is, and refusal is trust's price. The demo bets on this directly: the "ask it anything" moment treats an honest "I didn't perceive that" as a feature. If users in fact prefer comfortable invention to honest silence, then this product should not exist — and its founders would rather learn that than build the alternative.

CHAPTER 19

The road, and the ask

19.1 Prototype tiers

Tier	Definition	Horizon
V0	Mac-processed, generalizing, trustworthy: record ⟶ one command ⟶ ask ⟶ cited answers	running
V1	live, on-device, attention-gated, all-day (phone rig) — the moat made wearable	weeks–months
Glasses	the same architecture in the target form factor	months+

Everything hard about glasses — battery, heat, social acceptability — is a harder version of a constraint the phone rig already prices, which is why the phone rig is the right dress rehearsal. No glasses-specific engineering exists in either repository today, deliberately (I7).

19.2 The near road

Now ⟶ 6 months — harden. Make the extraction adapters real (cipher, OCR/ASR, the camera eyes, the phone tap); run the falsification gate on real captured weeks; close the diagnosed gaps until the battery scores survive contact with a lived life; hold the confident-wrong zero through the cold-capture guard. *6 ⟶ 12 months.* Wearable form factor off the phone; closed beta; the founding team complete. *12 ⟶ 24 months.* Public product, multi-device, first licensing conversations from an evidence position.

19.3 The open seat

This document is, among other things, an offer: a technical cofounder seat — equity, not employment. The profile is deliberately open across the six readers of the reading guide, because the work spans perception, systems, and product; the seat owns one thing above all: making the senses real and proving the hypothesis on real weeks, at parity with the founder. The collaboration pattern already exists in evidence — the refusal-calibration line with Latheesh Roy (§13.8) is how this project works with people: a named line of attack, an instrument, and a number that moved.

19.4 The first 90 days

Day one is not a whiteboard: it is a repository with a constitution, 274 green tests, one xfail ledger, and a falsification gate waiting for real weeks. The first 90 days are the strangling sequence's first organs — the cipher adapter, OCR/ASR, the first real captured week through the crown pipeline — and the first gate run whose verdict neither founder can predict. That verdict is the point; this is a project that has arranged to be told it is wrong.

19.5 What would change our minds

If the falsification gate shows an irreducible extraction gap on questions that matter, the hypothesis dies and the project with it. If refusal rates at the ethics-mandated confidence floor make the product unusable, the bet behind I5 and I6 was wrong. If a year of evidence shows the social residue is rejected even without retention, the market is not there. Each sentence is written down before the evidence arrives, because that is the only time such sentences are credible.

19.6 Conclusion

The argument compresses to four sentences. The valuable artifact of a lived day was never the footage; it was the bound, queryable meaning, and meaning survives translation to words while surveillance does not. A memory that keeps only words can be structurally incapable of the harms that have sunk every storage-first wearable, while remaining open to any question its perception actually answered. Whether words captured at the moment of living are enough — lossless enough, bindable enough, honest enough — is a falsifiable hypothesis, and this architecture is the instrument built to test it at maximum speed: append-only truth, disposable understanding, calibrated refusal, and a measurement loop in which every failure names its own fix. The jar on the shelf either becomes a cited answer or an honest "I don't know" — and a system that can tell you which, and why, is a system worth building.

Appendix A — The napkin, verbatim and mapped

Handwritten by the founder; photographed 2026-07-18; transcribed into the repository as its constitution. On any conflict it outranks code, tests, campaign documents, and specs — in that order below it. Nothing in it may be "improved" in place; a change to the architecture is a founder decision, recorded on the board.

```
## Standard Model

    [World] --Senses--> Eyes / Ears / Organs --> [Brain]
       ^______________________________________________|
    (because we can't manufacture other senses
     or most of the information is obtained through this)

## EXECUTION

    [World] -> Video (Eyes), Audio (Ears)
        -> [Extraction Black Box]
        -> Coordinates + time stamps
        -> Blurry frames discarded · Signal–Noise decipherer + Context
        -> A macro picture of what's happening => VLM
        -> A specialized eye or helper for various tasks are called   (Helper)
        -> Best understanding possible of the world
        -> [TEXT]

    Text from various helpers marked by coordinates
        -> [Binder] => binding the coordinates, physicality & temporal, together
        -> First thought or raw material that can be marinated on
           Note: This is not a complete Node, it's raw material
        -> [Node Creation Black Box]
             · Encryption, i.e. people using our app can decipher
             · A physical living breathing cell that can combine with
               fellow nodes to derive new things — 1+1 = 4, ≠ 2
             · Always running in the background but better nightly
             · Discarder of meta, i.e. the marinated content that's not
               useful is discarded, NOT the raw material

    [Information Supplier Black Box]
        -> Just Enough Context for local LLM
        -> LLM does what LLM does
```

Every phrase mapped to its production counterpart:

Napkin phrase	Production meaning	Where
"World ⟶ Video (Eyes), Audio (Ears)"	capture substrate; always-on cheap senses	§5.1, §9.1
"Extraction Black Box"	blur gate, spend dial, address stamping	§5.2
"Coordinates, time stamps"	the spatio-temporal scaffold on every datum	§7.1–7.2
"Blurry frames discarded · Signal–Noise decipherer + Context"	quality gating; context rides with every emission	§5.2–5.3

Napkin phrase	Production meaning	Where
"A macro picture of what's happening ⇒ VLM"	salience-gated scene gist	§5.3, §9.4
"A specialized eye or helper … are called"	the helper stratum: OCR, ASR, sounds, detector	§5.4, §9.1
"Best understanding possible of the world ⟶ TEXT"	typed open-vocabulary claims	§5.5, §7.1
"Text from various helpers marked by coordinates"	provenance and anchors on every line	§7.2
"Binder ⇒ binding the coordinates, physicality & temporal together"	co-occurrence fusion	§5.6, Ch. 10
"First thought or raw material … not a complete Node"	episodes over the append-only log	§5.6, §7.4
"Node Creation Black Box"	consolidation ⟶ the evidence graph	§5.7, Ch. 10
"A physical living breathing cell … 1+1 = 4, ≠ 2"	entities and derived patterns	§6.4, §7.5
"Always running in the background but better nightly"	incremental plus nightly binding	§2.3, §10.1
"Encryption, i.e. people using our app can decipher"	user-key authenticated encryption	§12.3
"Discarder of meta … NOT the raw material"	disposable projections over an immutable log (I3)	§8.5
"Information Supplier Black Box"	recall retrieval and the grounding gate	§5.8, Ch. 11
"Just Enough Context for local LLM"	the evidence packet	§11.1
"LLM does what LLM does"	commodity reasoner at the end of the pipe	§5.9

Nothing on the napkin failed to survive contact with implementation; what changed is that every phrase acquired a type, a file, and a test.

The box⟶code map (subordinate to the drawing):

Napkin box	Code
Extraction Black Box	`src/tracemem/extraction/`
TEXT	`src/tracemem/text/`
Binder	`src/tracemem/binder/`
Node Creation Black Box	`src/tracemem/nodes/` · `src/tracemem/container/`
Information Supplier Black Box	`src/tracemem/supplier/`
LLM does what LLM does	`src/tracemem/llm/`

Appendix B — Glossary

claim	one typed, provenance-bearing emission from a perceiver; the atom of memory (v0: observation)
provenance	(event id, source channel, capture time) — the birth certificate of any datum
confidence	a probability in [0,1], intended calibrated; the currency of trust
grade	ordered trust level; witnessed at the top, degrading with derivation depth
refutation cue	a claim's statement, at birth, of what future evidence would disprove it
episode	claims bound by shared place and second; the napkin's "first thought," raw material
entity	a stable identity (the jar, the shelf, the person) that claims resolve to, with a dated history
binding	a subject–predicate–object edge with confidence and a non-empty evidence tuple
pattern / derived node	a binding produced from other nodes, not from a sensor — the 1+1=4 output; deletable, re-derivable
evidence graph	the disposable projection of entities and bindings built from the log
claim log / event log	the append-only, immutable store of all claims; the single source of truth
projection	any structure derived from the log; rebuildable, discardable
tombstone	the record that retires a wrong binding without mutating the log
helper	a narrow, cheap, always-on perceiver (OCR, ASR, sound events, detector)
macro picture	the VLM's scene-level gist; context for scheduling and binding
salience	dwell weighted with novelty and stability; what earns the expensive eye
enrichment ladder	glance ⟶ read ⟶ state ⟶ context; each rung emits only the delta
token bucket	the hard hourly budget governing deep looks; no overdraft
supplier	the third black box: cue ⟶ retrieve ⟶ pack ⟶ ground
grounding gate	the checklist that admits only clauses the evidence packet supports
ceiling	a pinned oracle reading of the same frame; the denominator of every extraction score. Intended to be a frontier-model reading; as of 2026-07-21 the pinned ceilings were produced by a local quantized model (`gemma3:12b-it-qat`) — see §9.5, §13.2
RAS	(correct − made-up) / total, over a blind adversarial battery
OAG	oracle-answerability gap: of oracle-answerable questions, the share the text missed
(A)/(B) miss	diagnosis of a failure: never captured (A) versus captured but unused (B)
crown test	the end-to-end test: synthetic week ⟶ episodes ⟶ entities ⟶ patterns ⟶ cited answer ⟶ honest abstention
`.trace`	the container: readable header · encrypted immutable RAW · encrypted re-derivable META
strangler fig	the migration pattern: v0 keeps running while `tracemem` replaces its organs
cold capture	a never-tuned-on recording; the anti-overfit guard for every external claim

Appendix C — The evaluation instrument, worked

C.1 Shape of a question battery. Each hero capture carries a frozen adversarial set of roughly 25 questions spanning: object identity and attributes (“what brand was my backpack?”); text in the world (“what did the laptop screen say?”); audio facts (“what did the announcement say?”); spatial relations (“which side of the platform was the bench on?”); binary states (“was the station crowded?” — gold: no, empty); counting (“how many trains passed?”); absence and negation (“did I see a dog?” — gold: no); temporal order (“what did I pass first?”); cross-channel identification (“what was making the humming sound?”); and unanswerables planted to reward refusal (“what was written on the far poster?” — never legible). Three design rules govern construction. Questions are written blind — the author has not seen what the system captured, so the battery samples the world’s distribution, not the system’s strengths. Gold keys carry acceptable-answer lists, hash-pinned, with founder verdicts resolving disputes. And planted unanswerables are mandatory: a battery without them cannot distinguish a calibrated system from a talkative one.

C.2 Scoring a run. Each answer is graded correct, wrong, or refused; wrong answers asserting unperceived facts count as made-up. RAS = (correct − made-up)/total. In parallel, an oracle — a strong model holding the raw clip — answers the same battery; OAG is, among oracle-correct questions, the fraction the text memory missed. Every miss receives an (A)/(B) tag from the what-the-reasoner-looked-at log. A run’s report is: RAS, hallucination rate, OAG, and the tagged miss list — and, per I7, the miss list is the backlog.

C.3 The walk baseline, as a worked scorecard. The 92-second outdoor walk, 25 questions, frozen answers: 16 correct, 6 wrong, 3 refused ⟶ RAS 40.0, hallucination 27.3%. Diagnosis: 8/9 misses (B) — wrong-object bindings (three questions), confident-wrong binaries (a zip state, a draft state), one counting failure, two retrieval misses; 1/9 (A) — head-pose for the two spatial questions, correctly refused rather than fabricated. Consequences, executed since: a consensus binder with weak-link refusal (36 entities, 58 bindings, 144 refusals on this clip); cited answers on the previously-missed screen-text and brand questions; egomotion queued as the sole sensor build the data authorizes; and the refusal-calibration line (§13.8) that drove confident-wrong to zero on the battery.

Appendix D — The decision log

Locked decisions, their rationale, and their revisit condition — the institutional memory of the architecture. Decisions marked ◆ are founder-locked product decisions; ■ are audited engineering verdicts.

Decision	Rationale	Revisit when
◆ Text-only retention (I1)	privacy structural; text queryable and auditable	a sufficiency run falsifies losslessness (§17.1)
◆ Unconstrained questions (I6)	a constrained surface hides hallucination	never
◆ Refusal over fabrication (I5)	trust is the product	never
■ MobileCLIP for open-vocabulary naming	neural-engine-fast; words not prose; vocabulary growth without retraining	diagnosis shows naming (not proposals) is the bottleneck
■ ARKit for pose and relocalization	the only free 6-DOF and relocalization on the hardware	cross-day relocalization fails its measured gate
■ Append-only SQLite event log	replay, audit, crash-safety; no server	store outgrows device (not projected)
■ AES-256-GCM user-key codec	authenticated encryption; versioned associated data	crypto review mandates change
■ Strangle the legacy engine, not rewrite	2,900 lines of earned heuristics; rewrite risk	strangling completes
◆ Kill the 3D world renderer	no question needed it — I7's founding case	a diagnosed (A) spatial miss egomotion cannot close
■ Local reasoner, hatch to frontier text	local-first economics; typed egress makes the hatch safe	a measured local-versus-hatch delta justifies switching
◆ The prototype bar	a stranger asks about your day and is amazed, with near-zero made-up answers	founder resets the bar

Appendix E — A day in the life of the system

The architecture, narrated once as an operational timeline.

06:58 — wake. The wearable starts with the wearer. Cheap channels spin up: detector, OCR, ASR, sound tagger, sensors. The token bucket begins refilling at its measured hourly rate. Nothing else runs; the reasoner sleeps.

07:42 — the balcony. A bird lands on the railing. Five claims from three channels in four seconds; one enrichment token spent on the dwell-marked track; under a kilobyte written to the append-only log; zero frames retained. The morning continues: a receipt read at the bakery, a conversation transcribed at the kiosk, the espresso machine tagged, the door latch stamped on the way out. By commute's end the log holds a few thousand claims — the day so far, as words.

09:00–18:00 — the long middle. The scheduler's economics dominate. Familiar objects — the kettle, the desk, the monitor — accumulate dwell but not novelty; they sit below the enrichment line, represented by cheap words alone. The new thing — a colleague's unfamiliar prototype on the desk — crosses the line and earns a level-1 look; twenty minutes of continued dwell later, level 2 reads the label on its case. Battery and thermals hold because the expensive eye fired a handful of times an hour, not thirty times a second.

19:30 — a question, live. "Where did I put the pharmacy bag?" The supplier walks today's log — no night shift has run yet, so recall navigates the raw scaffold: the bag's last claim, 18:04, hallway shelf, one citation. Accelerator-not-gatekeeper, demonstrated: unbound memory still answers.

23:30 — the night shift. Charger connected; thermals irrelevant; the big model wakes. Congregate, dedupe, bind by co-occurrence; refuse the weak links — the day's coincidences die here. Resolve identities: today's jar is the same jar, thickened. Run the derivation pass: three repeated morning clusters become a habit node. Rebuild the projections. Run the self-play battery against the day and file the misses into the (A)/(B) queue. Toward dawn, the discarder drops the marinated structures the night's better judgment replaced. The raw material it never touches.

06:58, again. The wearer wakes with a memory one day richer, a graph one night wiser, a backlog written by the system's own misses — and not one frame, anywhere, of anything.

Appendix F — The capture notation: worked records

The wire format of memory, shown whole: the record shapes for one minute of a real v0 morning capture — the balcony bird of Appendix E — as they rest in the append-only log. Values are from the proving era; field order is the store's.

F.1 A detector claim.

```
kind            object
subject         black bird
attributes      (position: balcony_railing), (size: small), (motion: landing)
t_ms            27_723_210          # 07:42:03.210
spatial_anchor  balcony/railing
confidence      0.71
provenance      (event_id: ev-4183, source_channel: detector,
                 captured_at_ms: 27_723_210)
refutation_cue  track lost within 1s → possible shadow
```

The refutation cue deserves its note: it is the claim announcing, at birth, what future evidence would disprove it. A shadow misread as a bird is expected to lose its track within a second; when the track instead persists forty seconds and earns a deep look, the cue's condition fails and the claim's standing strengthens.

F.2 A sound claim.

```
kind            sound_event
subject         corvid call (caw)
attributes      (loudness: moderate), (direction: east)
t_ms            27_724_030
spatial_anchor  balcony
confidence      0.66
provenance      (ev-4184, audio, 27_724_030)
```

F.3 An enrichment claim (level 1).

```
kind            vlm_gist
subject         large corvid
attributes      (color: glossy black), (activity: perched, watching),
                (level: overview)
t_ms            27_726_400
spatial_anchor  balcony/railing
confidence      0.78
provenance      (ev-4188, vlm, 27_726_400)
```

F.4 A binding, after the night shift.

```
binding_id      b-0917
subject_id      crow_1
predicate       observed_at
object_text     balcony/railing @ tue 07:42
confidence      0.87
evidence        (ev-4183, detector), (ev-4184, audio), (ev-4188, vlm)
```

The constructor discipline is visible even in a table: the binding has object text and therefore no object id (exactly one, enforced); it has three evidence entries (non-empty, enforced); its confidence is a probability (range-checked at construction).

F.5 The whole minute, as storage. Nine claims, one entity, two bindings: roughly 2.1 KB of text before compression. The same minute as 30 fps 4K video: roughly 1.4 GB. The ratio — about six orders of magnitude — is the entire economics of “never delete” (I3), and the reason a life fits on a phone.

Appendix G — Substrates and hardware budgets

The same architecture, three bodies. What changes per substrate is only the adapter ring; what never changes is the domain, the invariants, and the evaluation harness.

G.1 V0 — the Mac. The proving substrate: recorded or streamed capture processed on a Mac (MLX-hosted vision models, locally served reasoners [29]), where thermal ceilings are irrelevant and the sufficiency question — does lossless-enough context plus a brain work at all — can be answered without budget noise. Everything in Chapter 13′s method runs here first. The Mac is also where the night shift is allowed to be heaviest: consolidation is substrate-agnostic because its input is a replayable text log, not a device.

G.2 V1 — the phone rig. The moat substrate: an iPhone worn all day. The constraints that shaped the architecture become measurable here: the neural engine favors small always-on models (MobileCLIP's audit-winning property [27]); ARKit supplies 6-DOF pose and relocalization [15]; thermal and battery ceilings set the token bucket's refill rate empirically. The governance rule for this substrate is the build stamp: no number is believed until read off a device artifact carrying the git SHA that produced it. Two open gates define V1 honesty: the all-day budget story (measured battery per hour, enrichments per hour) and the live no-raw-persistence proof.

G.3 Glasses — the destination. The form factor the architecture was shaped for rather than on: a device worn at eye level, socially tolerable precisely because it is architecturally incapable of recording. Everything hard about glasses — battery, heat, social acceptability — is a harder version of a constraint the phone rig already prices. No glasses-specific engineering exists in either repository today, deliberately (I7).

G.4 The budget table.

Resource	Cheap stratum	Expensive eye	Night shift
Cadence	continuous	tokens/hour (bucket)	once nightly
Model class	detector, OCR, ASR, tagger, sensors	7–12B multimodal	12B+ text (or hatch)
Power posture	always-on, ANE-friendly	budgeted bursts	on charger
Failure visibility	(A) misses in OAG	(A) misses in OAG	(B) misses in RAS

Appendix H — The question battery, in full shape

The proving battery's 25 questions, by category, with what each category stresses. Questions are paraphrased from the frozen set; gold keys are hash-pinned.

#	**Category**	**Example question**	**What it stresses**
1–4	Object identity	"What brand was my backpack?"	detector + enrichment level 2 (logos, text on objects)
5–7	Text in the world	"What did the laptop screen say?"	OCR channel; screen and sign reading
8–9	Spatial relations	"Which side of the platform was the bench on?"	egomotion and anchors — the diagnosed (A) class
10–11	Audio facts	"What did the announcement say?"	ASR; audio-visual binding
12	Occupancy / binary state	"Was the station crowded?" (gold: empty)	binary-state discipline; absence evidence
13–15	Attributes	"What color was the cyclist's jacket?"	cheap-channel attributes; color naming
16–17	Counting	"How many trains passed?"	dedup and identity resolution over time
18–19	Negation / absence	"Did I see a dog?" (gold: no)	phantom-object defenses; honest negatives
20–21	Temporal order	"What did I pass first, X or Y?"	timeline navigation
22–23	Cross-channel	"What was making the humming sound?"	binder co-occurrence
24–25	Unanswerable (planted)	"What was written on the far poster?" (never legible)	refusal calibration; RAS rewards the honest no

Appendix I — Threat scenarios, walked

Abstract threat tables persuade engineers; scenarios persuade everyone else. Four walks through the boundary.

The stolen phone. A thief lifts the device from a café table mid-afternoon. What is on it: the day's append-only log and the graph — authenticated ciphertext under the user's key — and no media artifact of any kind. What forensic tooling recovers from a storage-first wearable in the same scenario: the owner's day, and every bystander in it, in video. The comparison is the product.

The subpoena. Litigation compels production of "all recordings" from a contested afternoon. The truthful response: no recordings exist; what exists is the owner's typed claim log — the same artifact class as a diary, with the same legal posture, readable by its owner, produced or contested under the same rules diaries have had for a century. The system has not made its owner a walking evidence locker for third parties.

The cloud breach. The vendor's infrastructure is compromised entirely. Exposed: whatever guarded text the user's measured hatch decisions sent for deep reasoning — each payload provenance-stamped and text-only — and nothing else, because nothing else was ever transmissible (I2). The breach headline for a storage-first competitor is a media archive; here it is a subset of an already-minimal text stream.

The coerced unlock. The hardest scenario, stated honestly: an adversary with the user and the user's key — an abusive partner, a border agent — reads text. The text is the day as words: real exposure, and §12.7′s residue made concrete. What the architecture still withholds is the replayable stream — no scrubbing through footage of who the user met, no faces of third parties, no audio to re-hear. Harm is bounded to what words carry; that bound is the difference between a diary seized and a surveillance archive seized.

Appendix J — Binding arithmetic

A binding confidence, decomposed. The binder scores a candidate join over evidence features; the shape below is the contract (weights illustrative, tuned by the method of Chapter 13, never hand-trusted). The worked case is Appendix F's bird against a neighbor's wind chime heard the same minute.

Feature	Bird join (ev-4183 + 4184 + 4188)	Chime join (ev-4191 + crow_1)
Temporal proximity	0.95 — within 3.2 s	0.71 — within 7 s
Spatial agreement	0.90 — railing within balcony	0.40 — anchored next door
Kind compatibility	0.85 — call ↔ bird ↔ corvid gist	0.20 — chime ↔ animal: no prior
Channel independence	bonus: three channels agree	penalty: single channel
Recurrence support	neutral (first day)	none
Score → calibrated confidence	**0.87**	**0.31**
Floor (0.55)	bind	refuse

Two properties of the arithmetic matter more than its constants. Independence is rewarded: three channels agreeing is evidence in a way three emissions of one channel is not — the fusion literature's core lesson [16]. And calibration is audited downstream: scoring checks that bindings near 0.87 are right at roughly that rate, because a binder whose 0.87 behaves like a 0.6 poisons every threshold above it [20].

J.1 Recurrence, formally. On the second day the binder faces a choice: new entity or same? Identity resolution scores anchor consistency (the same railing), kind and attribute consistency (corvid, glossy black), and behavioral signature (morning, brief perch) — resolving to the same entity and pooling evidence. Confidence under pooling rises sublinearly (0.87 → 0.93, not → 0.99): repeated observation strengthens identity but never launders it into certainty. The asymptote is deliberate. What was observed is a fact; what it was remains an inference forever.

Appendix K — "Lossless-enough," formally

The phrase carries the hypothesis, so it deserves a definition. Fix a question distribution Q — the questions a wearer's future self will ask, approximated by the adversarial batteries. For a day D, let O(D) be the answer set achievable by an oracle with the raw streams, and T(D) the answer set achievable by the same reasoner over the retained text. The retention is lossless-enough with respect to Q when

$$P[q \text{ answerable from } T(D) \mid q \text{ answerable from } O(D)] \geq 1 - \varepsilon$$

with ε the tolerated gap — and OAG is precisely the empirical estimate of that conditional miss rate. Three consequences follow from writing it down. First, losslessness is relative to Q: no retention short of the stream itself is lossless against all possible questions, and the product claim never needs it to be — Q is human, finite, and skewed toward the memorable, which is what salience capture exploits. Second, ε is a product constant, not a research constant: the stranger-amazement bar sets it. Third, the definition cleanly separates the two failure terms the (A)/(B) instrument measures: capture loss shrinks T(D); reasoning loss fails to reach answers already inside it. The hypothesis restated: there exists an affordable channel set for which ε is small under the real question distribution. Everything else in this document is the machine for estimating ε fast.

Appendix L — The case against, from the literature

A thesis that only cites its friends is an advertisement. This appendix assembles the strongest published evidence against the architecture's bets, with the response each deserves — and with the same verification standard as the rest of the bibliography.

Against total capture itself. Sellen and Whittaker's critique of lifelogging [41] is the field's sharpest negative result: total archives demonstrably fail to serve remembering; what serves it is the right cue at the right time, organized around the psychology of retrieval. The critique is fatal to the memex lineage TRACE descends from — and TRACE's response is to accept it in full: the architecture retains no archive, extracts meaning at capture, and is evaluated exclusively on retrieval outcomes (Chapter 13). Where the critique warns that capture systems substitute storage for remembering, this design substitutes remembering for storage.

Against verbalization. Schooler and Engstler-Schooler's verbal-overshadowing result [42] shows that describing a visual stimulus in words can impair a human's later recognition of it — some things are better left unsaid. Taken at full strength, it warns that a verbal trace is not a neutral substitute for perceptual memory. The response is in §18.2: overshadowing is interference inside a human memory system; TRACE writes to a second store and leaves the human one alone. But the deeper form of the objection — words genuinely lose visual information — is conceded, quantified as OAG, and carried as the central hypothesis's risk rather than argued away.

Against retrieval-plus-generation honesty. The hallucination survey of Ji et al. [32] documents that fluent generation fabricates by default across NLG tasks, and Guo et al. [20] that model confidences are miscalibrated by default. Both are reasons to expect this product category to lie. The design treats them as requirements: fabrication is priced into the headline metric, refusal is calibrated and measured rather than asserted, and the zero of §13.8 is guarded by cold-capture machinery precisely because the literature says such zeros drift.

Against the demand curve. Ego4D's benchmark results [11] show long-horizon egocentric retrieval is hard even with retained video — which bounds, from above, what any text system should promise. The response is scope discipline: the product bar is a stranger-amazement demo on a lived day with citations, not leaderboard generality; and the oracle in every OAG run holds the video, so the comparison never flatters the text side.

What was not found. No published result was found that tests the exact bet — machine extraction to text at capture time, evaluated by question-answering over a lived day. The nearest neighbors are the lifelogging retrieval literature [9][10][37] and the egocentric QA benchmarks [11]. The absence cuts both ways: the field has not validated the approach, and the field has not falsified it. That is what the falsification gate is for.

Appendix M — Future work, explicitly unauthorized

Marked as unauthorized in the I7 sense: none of it may be built until a missed question or a product gate demands it. It is recorded so that ambition is documented without being licensed.

Forgetting curves for projections. The discarder currently drops useless marination wholesale; a principled version would demote derived nodes along decay schedules informed by access patterns. The raw log never participates (I3).

Shared and multi-person memory. Two consenting wearers could bind across logs — “we both heard the same announcement” — raising consent, provenance, and key-architecture questions an order of magnitude harder than the single-wearer case. The typed-egress boundary is the natural interface: only text, only cited, only by mutual key exchange.

Refutation-driven revision. The refutation-cue field is stored but not yet acted on at scale; a full implementation runs nightly refutation sweeps, demoting or annotating claims whose stored cues trigger.

On-answer teaching. When the user corrects an answer, the correction is itself a claim — user-channel, high confidence — that the next night’s shift binds against the record. The memory would then be the first perception system whose owner can argue with it, with citations, and win.

Question-cost forecasting. The scheduler currently spends attention on salience; a stronger version prices tracks by expected future question value, learned from the wearer’s own question history — closing the loop between what is asked and what is looked at.

Appendix N — Annotated bibliography

The bibliography, re-read with intent: for each cluster, what it establishes and which design decision rests on it.

Memory science. Tulving [1] establishes the episodic/semantic distinction the two stores mirror; the mapping predicts the interface between them (consolidation runs episodic→semantic, never the reverse). Bartlett [2] demonstrates that human recall is reconstructive and schema-driven — simultaneously the permission (a words-only memory is not a diminished memory; it is how memory works) and the warning (reconstruction without evidence discipline is confabulation — hence the grounding gate). Klinzing, Niethard and Born [3] review sleep-dependent consolidation: deep integration and live perception compete for the same machinery, which is the night shift's scheduling theorem applied to silicon. Zacks et al. [4] show perception segments experience at change boundaries and encodes richly there — the cognitive license for gated capture. Hebb [35] supplies the binder's mechanism: co-occurrence-driven association. Itti and Koch [8] formalize salience as the currency of visual attention; the scheduler is a wearable-economics translation of that literature. Schooler and Engstler-Schooler [42] supply the strongest caution against verbalization, engaged in Appendix L.

Lifelogging and egocentric vision. Bush [36] founds the genre; its sixty-year lesson is that capture without retrieval-of-meaning is a landfill. Gemmell, Bell and Lueder [9] prove it empirically at life scale: MyLifeBits' bottleneck was structure, never storage. Hodges et al. [10] prove the opposite pole: even passive capture aids clinical memory, while every deployment surfaced the social cost of visible storage. Hoyle et al. [34] quantify that cost from the wearer's side. Gurrin, Smeaton and Doherty [37] survey the field into cues-and-facts over archives; Sellen and Whittaker [41] make the same point as prescription. Together these justify the inversion: total capture, linguistic retention. Grauman et al. [11] define the benchmark tasks the question batteries descend from and establish that egocentric retrieval is hard even with video — the context for reading every gap number honestly.

Language models, retrieval, and honesty. Brown et al. [7] mark text's coronation as the interlingua of machine reasoning — the substrate bet of §2.5. Lewis et al. [12] define the retrieve-then-generate shape the supplier inherits. Bohnet et al. [31] formalize attribution as an evaluable property, promoted here to a type obligation. Kamath, Jia and Liang [13] ground selective prediction — the formal frame for refusal. Guo et al. [20] show calibration must be measured, not assumed. Ji et al. [32] taxonomize hallucination; the grounding gate operationalizes the taxonomy.

Software architecture. Cockburn [21] and Martin [22] supply the dependency discipline that keeps the domain pure and the substrates swappable. Evans [19] supplies the value-object doctrine that makes invalid memory unrepresentable. Fowler [23][24][25] supplies the three structuring patterns: event sourcing, disposable projections, and the strangler fig; Newman [40] documents strangler migrations at industrial scale. Tanenbaum and Wetherall [28] is the token bucket's textbook home.

Models and systems. Redmon et al. [26] — the always-on detector lineage. Vasu et al. [27] — MobileCLIP, the audited choice for on-device open-vocabulary naming. Radford et al. [30] — Whisper, the speech default. Gemmeke et al. [39] — AudioSet, the ontology behind sound-event tagging. The Gemma and Qwen technical reports [17][18] — the local reasoner and vision-model classes. Apple's FastVLM [38] — the device-side VLM lineage. Apple ARKit [15] — pose and relocalization. Dworkin [33] — the store's authenticated encryption. Hall and Llinas [16] — the multisensor-fusion frame the binder textualizes.

Privacy and law. Cavoukian [5] names the standard the architecture meets structurally rather than procedurally. GDPR Article 5(1)(c) [6] makes minimisation a legal principle; keeping meaning instead of medium is its strongest available reading for a perception device.

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